

Operating Systems

Quiz 3

Q1:

Implement a multi-threaded ticket booking system where multiple customers (threads) try to book tickets concurrently. The system should ensure that:

1. No two threads book the same ticket (avoid race conditions).
2. Only available tickets can be booked (prevent overbooking).
3. The booking process is thread-safe using mutex locks.

Details:

- Use an array to store available tickets.
- Each thread represents a customer trying to book a ticket.
- Use a mutex (`pthread_mutex_t`) to ensure mutual exclusion.
- If tickets are available, the customer books one ticket and updates the system.
- If no tickets are left, the customer is informed that booking is not possible.

Test your program with 15 customers (threads) and 10 tickets.